

1. Consider the function $h(x) = \int_0^x t^2 dt$.

a) Fill out the following table of values for the function $h(x)$

x	-3	-1	0	1	3
$h(x)$					

b) Provide a rough sketch of what the function $h(x)$ looks like.

c) Determine the value of $h'(2)$.

2. Find the derivative of the function $g(x) = \int_x^8 [e^t - \ln(t^2 + 1)] dt$.

3. Find the derivative of the function $g(x) = \int_4^{\ln(x^2+1)} \left[\frac{\sin(t) + 2}{e^t + t} \right] dt$.

4. Compute $\frac{d}{ds} \left[\int_{\cos(s)}^{s^2+2} \frac{1}{x^2+1} dx \right]$

5. Let $g(x) = \int_x^{x^2} 6 dt$.

a) Use your calculator to graph this function on the interval $[-1, 1]$.

b) Compute $g'(x)$ by first evaluating the integral using the FTC2 and then differentiating.

c) Compute $g'(x)$ by using the FTC1 directly. Note that you will have to rewrite the function a bit first.

d) Show that the two results from parts (b) and (c) are consistent.

6. Determine the equation of the tangent line to $g(x) = \int_0^{x^2+x} t dt$ at $x = 3$.

7. Let $F(x) = \int_0^x \frac{t-3}{t^2+7} dt$.

- a) Find all intervals where $F(x)$ is increasing.
- b) Find all intervals where $F(x)$ is concave up.

8. Compute the maximum of the function $g(x) = \int_{e^x}^{2e^x} \frac{1}{1+w^2} dw$.

9. Find the maximum value of the area of the region under the curve $y = 4x - x^3$ from $x = a$ to $x = a+1$, for all $a > 0$.

10. If $x \sin(\pi x) = \int_0^{x^2} f(t) dt$, where f is a continuous function, find $f(4)$.